

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA0713 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Dr.V.Parthipan

Code of Conduct:

*I **Shane Rooney . A[192511057]** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Shane Rooney . A

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte IP** packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted,

and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1:

A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport Layer and Data Link Layer ensure reliable communication.

Question 1 – Solution

Given Data

<i>Parameter</i>	<i>Value</i>
File Size	150 MB
Bandwidth	50 Mbps
Segment Size	1500 Bytes
Data Link Layer Overhead	40 Bytes per Frame

Step 1: Convert File Size into Bytes

Since

1 MB = 1,000,000 Bytes

Therefore,

$$150 \times 1,000,000 = 150,000,000 \text{ Bytes}$$

Total File Size = 150,000,000 Bytes

Step 2: Calculate Number of Segments

Formula

$$\text{Number of Segments} = \frac{\text{Total File Size}}{\text{Segment Size}}$$

Substitute the values

$$\begin{aligned} &= \frac{150,000,000}{1500} \\ &= 100,000 \end{aligned}$$

Answer

$$\text{Total Number of Segments} = 100,000$$

Step 3: Number of Frames

*Each Transport Layer segment is encapsulated into one Data Link Layer frame.
Therefore,*

$$\text{Frames} = \text{Segments}$$

Hence,

$$\text{Total Number of Frames} = 100,000$$

Step 4: Calculate Total Data Link Layer Overhead

Each frame contains

- *Header + Trailer = 40 Bytes*

Total Frames

$$= 100,000$$

Formula

$$\begin{aligned} \text{Total Overhead} &= 100,000 \times 40 \\ &= 4,000,000 \text{ Bytes} \end{aligned}$$

Convert into MB

$$= \frac{4,000,000}{1,000,000} = 4 \text{ MB}$$

Answer

$$\text{Total Overhead} = 4,000,000 \text{ Bytes (4 MB)}$$

Step 5: Calculate Total Data Actually Transmitted

Original File

= 150 MB

Added Overhead

= 4 MB

Therefore

$$150 + 4 = 154 \text{ MB}$$

Convert into bits

$$154 \times 8 = 1232 \text{ Megabits}$$

Step 6: Calculate Transmission Time

Formula

$\text{Transmission Time} = \frac{\text{Total Data (Megabits)}}{\text{Bandwidth (Mbps)}}$

Substitute values

$$\begin{aligned} &= \frac{1232}{50} \\ &= 24.64 \text{ seconds} \end{aligned}$$

Answer

Estimated Transmission Time = 24.64 seconds

Final Calculated Answers

<i>Calculation</i>	<i>Answer</i>
<i>File Size</i>	<i>150 MB</i>
<i>Number of Segments</i>	<i>100,000</i>
<i>Number of Frames</i>	<i>100,000</i>
<i>Overhead per Frame</i>	<i>40 Bytes</i>
<i>Total Overhead</i>	<i>4,000,000 Bytes (4 MB)</i>
<i>Total Data Sent</i>	<i>154 MB</i>
<i>Estimated Transmission Time</i>	<i>24.64 seconds</i>

Explanation of the Transport Layer

The Transport Layer (Layer 4) is responsible for providing reliable communication between the sender and receiver. Instead of sending the entire file as one large block, it divides the data into smaller units called segments. In this problem, each segment contains 1500 bytes of user data.

The Transport Layer performs several important functions:

- *Segmentation: Divides large files into manageable segments.*
- *Sequencing: Assigns sequence numbers so segments can be reassembled correctly at the destination.*
- *Error Recovery: Retransmits lost or damaged segments.*
- *Flow Control: Prevents the sender from overwhelming the receiver.*
- *End-to-End Communication: Ensures reliable data transfer between applications on different devices.*

Because of these functions, even if some segments are lost or arrive out of order, the Transport Layer can detect the problem and recover the missing information.

Explanation of the Data Link Layer

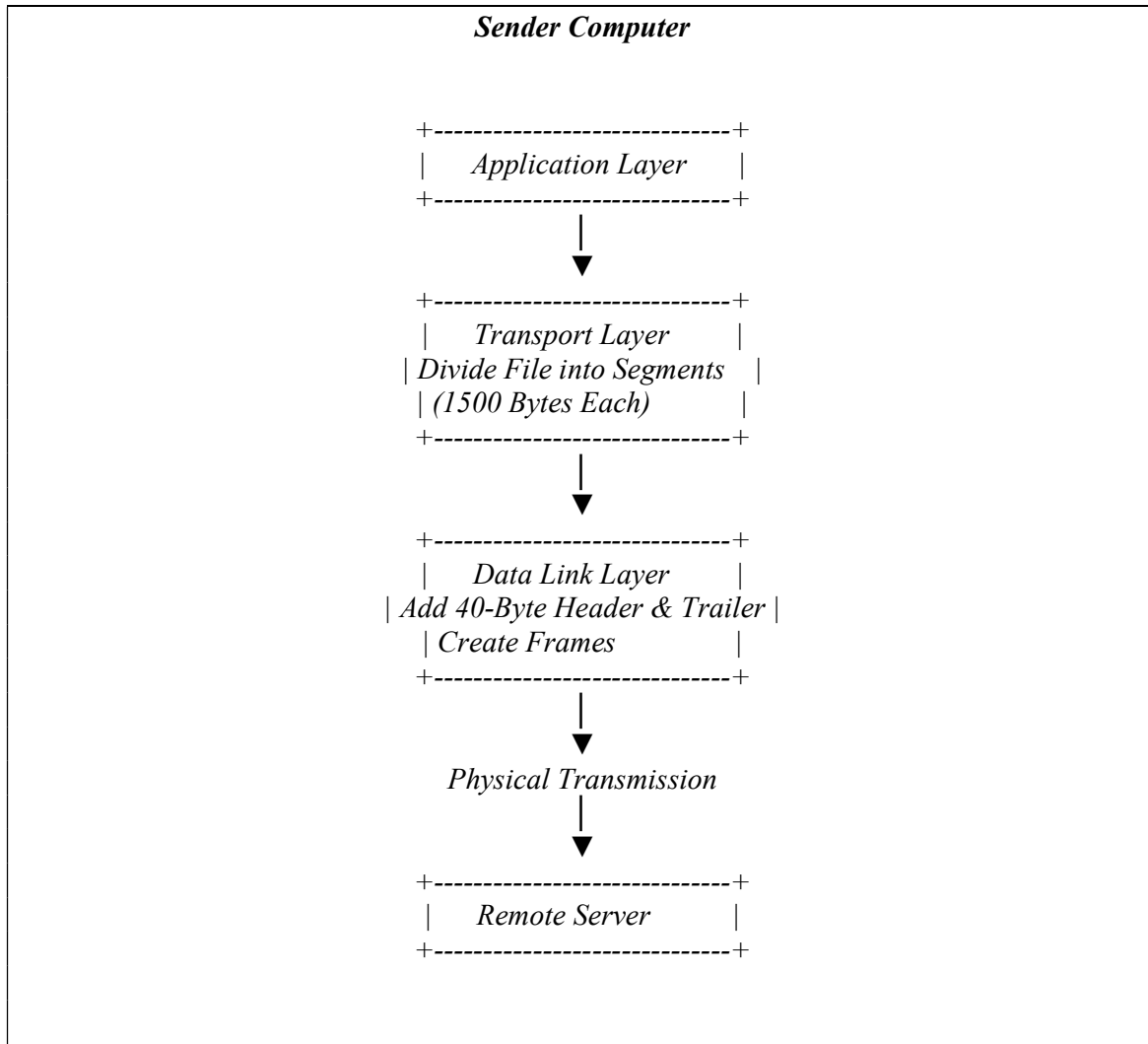
The Data Link Layer (Layer 2) is responsible for reliable communication between two directly connected devices on the same network. After receiving segments from the Transport Layer, it encapsulates each segment into a frame by adding a header and a trailer. In this question, the total size of the header and trailer is 40 bytes.

The Data Link Layer performs the following tasks:

- *Framing: Encapsulates segments into frames.*
- *Physical Addressing: Adds MAC addresses of the sender and receiver.*
- *Error Detection: Uses mechanisms such as a Frame Check Sequence (FCS) or CRC to detect transmission errors.*
- *Flow Control: Regulates the rate of frame transmission between directly connected devices.*
- *Media Access Control: Controls how multiple devices share the communication medium.*

These functions ensure that each frame reaches the next network device accurately and without corruption.

How the Transport Layer and Data Link Layer Work Together



Question:2

A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given Data

- *Sliding Window Size = 6 Frames*
- *Total Frames = 48*
- *Frame Size = 1024 Bytes*
- *Lost Frames = 1 per Window*

Step 1: Number of Transmission Windows

Formula

$$\begin{aligned} \text{Number of Windows} &= \frac{\text{Total Frames}}{\text{Window Size}} \\ &= \frac{48}{6} = 8 \end{aligned}$$

Answer: 8 Windows

Step 2: Total Retransmissions

One frame is lost in every window.

$$\text{Retransmissions} = 8 \times 1 = 8$$

Answer: 8 Frames

Final Answers

<i>Calculation</i>	<i>Answer</i>
<i>Number of Windows</i>	<i>8</i>
<i>Total Retransmissions</i>	<i>8 Frames</i>

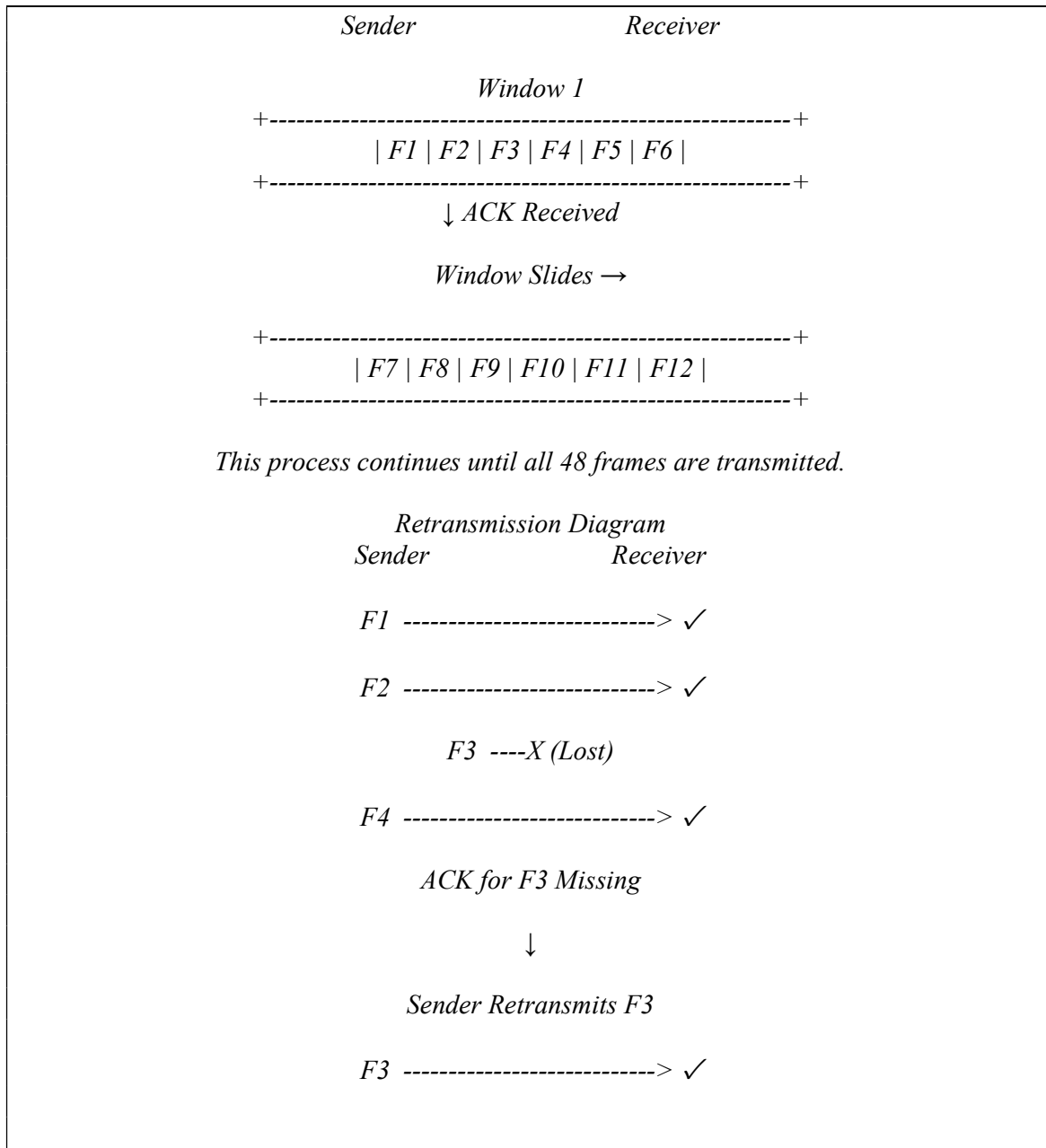
Flow Control

Flow control is a Transport Layer mechanism that controls the rate of data transmission between the sender and receiver.

Advantages:

- *Prevents receiver overload.*
- *Reduces packet loss.*
- *Improves data transmission efficiency.*
- *Ensures reliable communication.*
- *Increases network throughput.*

Sliding Window Diagram



Question:3

A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given Data

- IP Packet Size = 12,000 Bytes
- MTU = 1500 Bytes
- IP Header Size = 20 Bytes

Step 1: Calculate Maximum Data per Fragment

Since each fragment requires a 20-byte IP header,

$$\text{Data per Fragment} = 1500 - 20 = 1480 \text{ Bytes}$$

Answer: 1480 Bytes

Step 2: Calculate Number of Fragments

Formula

$$\begin{aligned} \text{Number of Fragments} &= \left\lceil \frac{12000}{1480} \right\rceil \\ &= \lceil 8.108 \rceil = 9 \end{aligned}$$

Answer: 9 Fragments

Step 3: Calculate Total Header Overhead

Each fragment has a 20-byte IP header.

Formula

$$\text{Total Header Overhead} = 9 \times 20 = 180 \text{ Bytes}$$

Answer: 180 Bytes

Final Answers

Calculation	Answer
Data per Fragment	1480 Bytes
Number of Fragments	9
Header per Fragment	20 Bytes
Total Header Overhead	180 Bytes

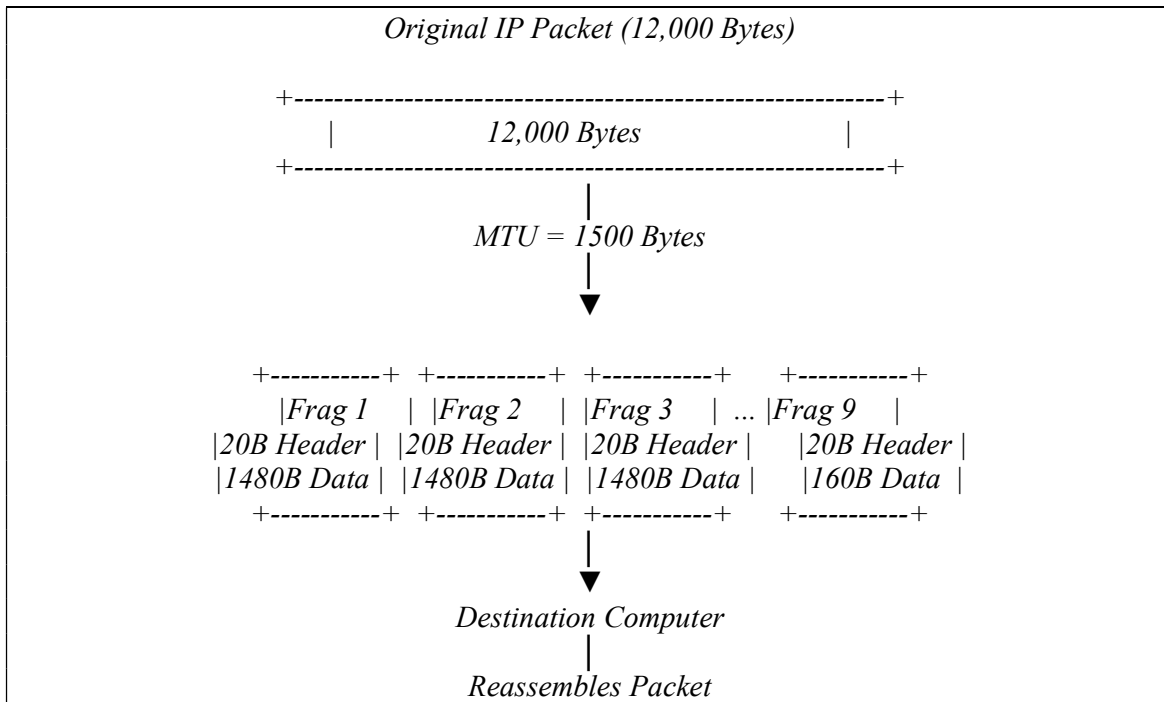
Explanation

The Network Layer (Layer 3) is responsible for delivering packets from the source to the destination across different networks.

Its main functions are:

- *Packet Fragmentation: Splits large packets into smaller fragments when the packet size exceeds the MTU.*
- *Logical Addressing: Uses IP addresses to identify the sender and receiver.*
- *Routing: Finds the best path to reach the destination.*
- *Reassembly: The destination device combines all fragments to reconstruct the original packet.*
- *Reliable Delivery: Uses fragment identification, fragment offset, and the More Fragments (MF) flag to ensure correct reassembly.*

IP Fragmentation Diagram



Question:4

A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the **number of transmission windows required**. If **one frame in every window is lost** and must be retransmitted, determine the **total number of retransmissions**. Explain how **flow control** improves communication efficiency. (This question is identical to Question 2.)

Given Data

- Sliding Window Size = **6 Frames**
- Total Frames = **48**
- Frame Size = **1024 Bytes**
- Lost Frames = **1 per Window**

Step 1: Calculate Number of Transmission Windows

Formula

$$\begin{aligned} \text{Number of Windows} &= \frac{\text{Total Frames}}{\text{Window Size}} \\ &= \frac{48}{6} = 8 \end{aligned}$$

Answer: 8 Windows

Step 2: Calculate Total Retransmissions

Formula

$$\begin{aligned} \text{Retransmissions} &= \text{Number of Windows} \times 1 \\ &= 8 \times 1 = 8 \end{aligned}$$

Answer: 8 Retransmissions

Final Answers

Calculation	Answer
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Number of Windows	8
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Total Retransmissions	8 Frames
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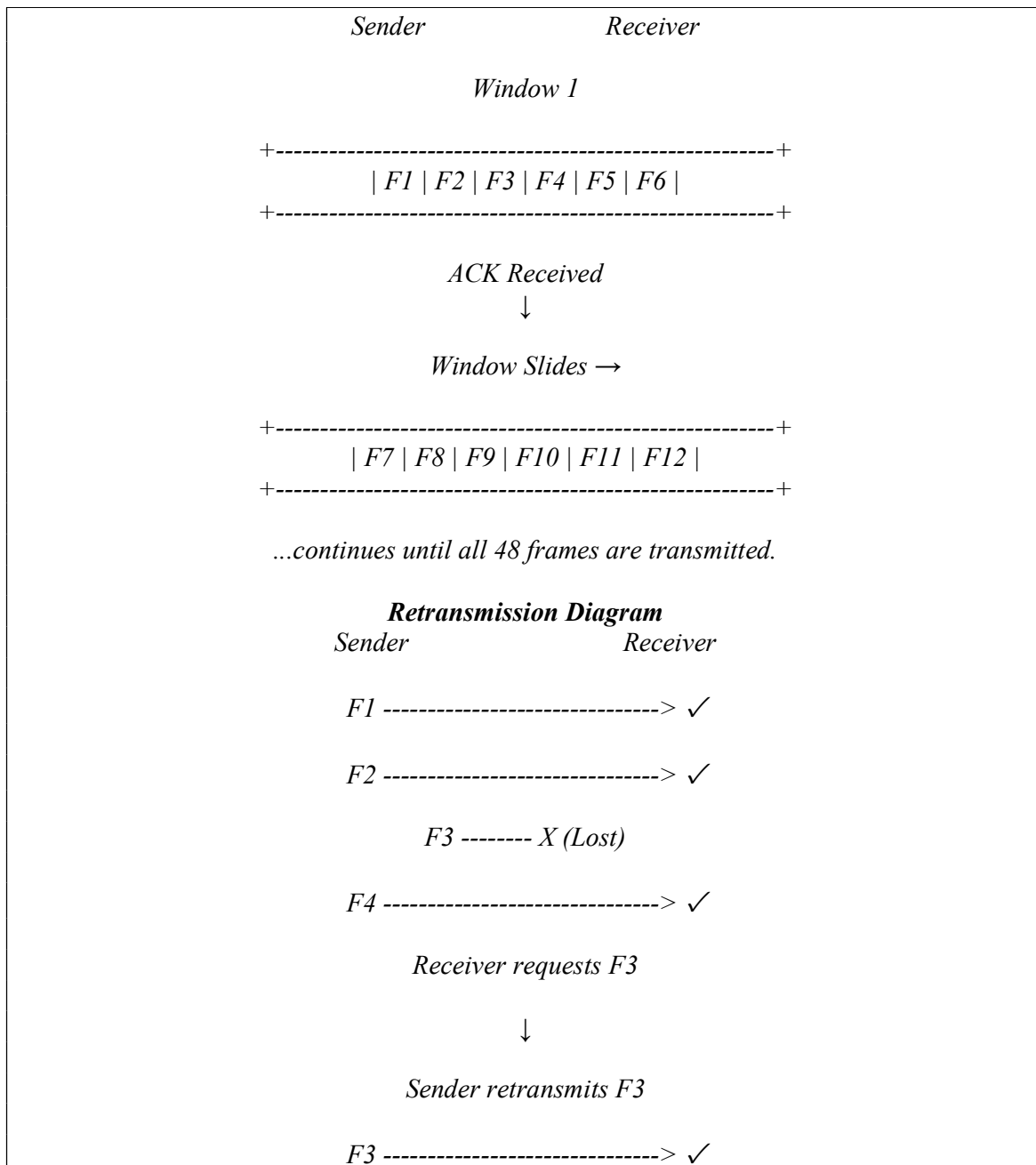
Flow Control

Flow control is a Transport Layer technique that regulates the speed of data transmission between the sender and receiver.

Advantages:

- Prevents receiver buffer overflow.
- Reduces packet loss.
- Improves network efficiency.
- Increases throughput.
- Ensures reliable communication.

Sliding Window Diagram



Question:5

A multimedia application transfers a **600 MB** video file. The **Session Layer** inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the **number of checkpoints created before failure** and calculate the **amount of data that must be retransmitted after recovery**. Explain the importance of synchronization in the **Session Layer**.

Given Data

- Total File Size = **600 MB**
- Checkpoint Interval = **60 MB**
- Failure Occurs After = **390 MB**

Step 1: Calculate Number of Checkpoints

Formula

$$\begin{aligned} \text{Number of Checkpoints} &= \frac{390}{60} \\ &= [6.5] = 6 \end{aligned}$$

Answer: 6 Checkpoints

Step 2: Find Last Checkpoint

Checkpoints are created at:

- 60 MB
- 120 MB
- 180 MB
- 240 MB
- 300 MB
- 360 MB

The failure occurs at **390 MB**, so the **last saved checkpoint** is at **360 MB**.

Step 3: Calculate Data to be Retransmitted

Formula

$$\begin{aligned} \text{Retransmitted Data} &= 390 - 360 \\ &= 30 \text{ MB} \end{aligned}$$

Answer: 30 MB

Final Answers

Calculation	Answer
Number of Checkpoints	6
Last Checkpoint	360 MB
Failure Occurs At	390 MB
Data to be Retransmitted	30 MB

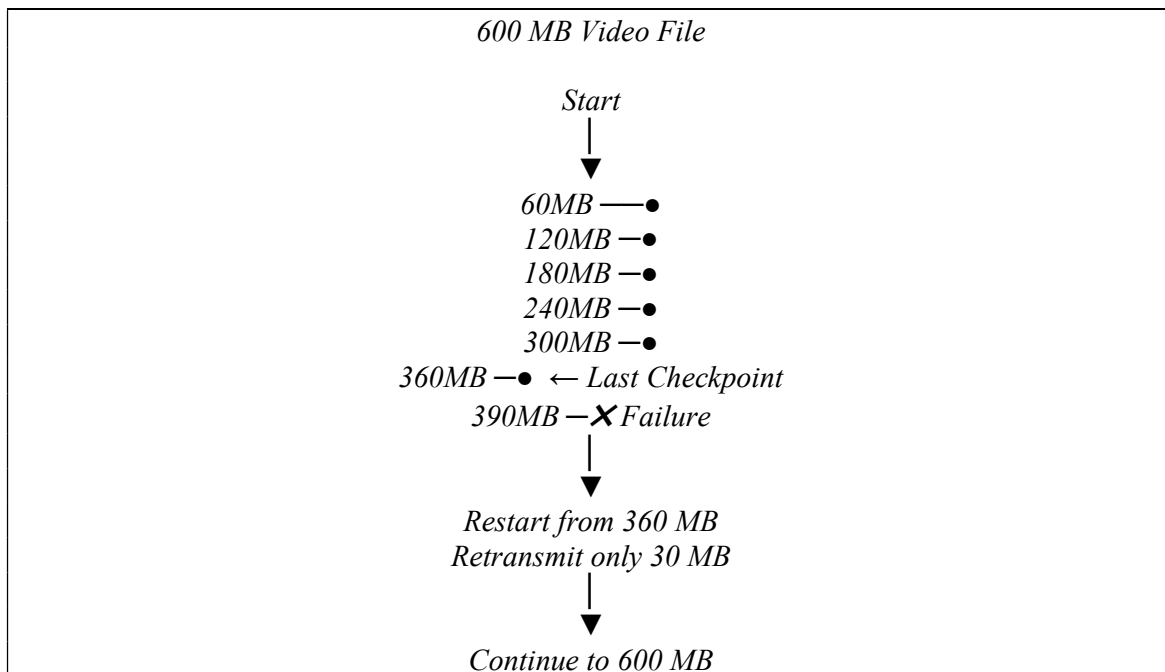
Importance of Synchronization in the Session Layer

The **Session Layer (Layer 5)** manages communication sessions between two devices. It inserts **synchronization checkpoints** during data transfer.

Advantages

- *Prevents retransmission of the entire file.*
- *Saves transmission time.*
- *Reduces network traffic.*
- *Enables recovery from failures.*
- *Improves communication reliability.*

Session Layer Synchronization Diagram



Question:6

Before transmission, the **Presentation Layer** compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5% overhead** to the compressed size. Calculate the **compressed file size**, the **encrypted file size**, and the **total percentage reduction** compared to the original file. Explain how the **Presentation Layer** improves transmission efficiency and security.

Given Data

- Original File Size = **900 MB**
- Compression = **40%**
- Encryption Overhead = **5%**

Step 1: Calculate Compressed File Size

Compression reduces the file size by **40%**.

Formula

$$\begin{aligned}\text{Compressed Size} &= 900 \times (1 - 0.40) \\ &= 900 \times 0.60 \\ &= 540 \text{ MB}\end{aligned}$$

Answer: 540 MB

Step 2: Calculate Encrypted File Size

Encryption adds **5% overhead** to the compressed file.

Formula

$$\begin{aligned}\text{Encrypted Size} &= 540 + (540 \times 0.05) \\ &= 540 + 27 \\ &= 567 \text{ MB}\end{aligned}$$

Answer: 567 MB

Step 3: Calculate Total Percentage Reduction

Formula

$$\begin{aligned}\text{Reduction} &= \frac{900 - 567}{900} \times 100 \\ &= \frac{333}{900} \times 100 \\ &= 37\%\end{aligned}$$

Answer: 37% Reduction

Final Answers

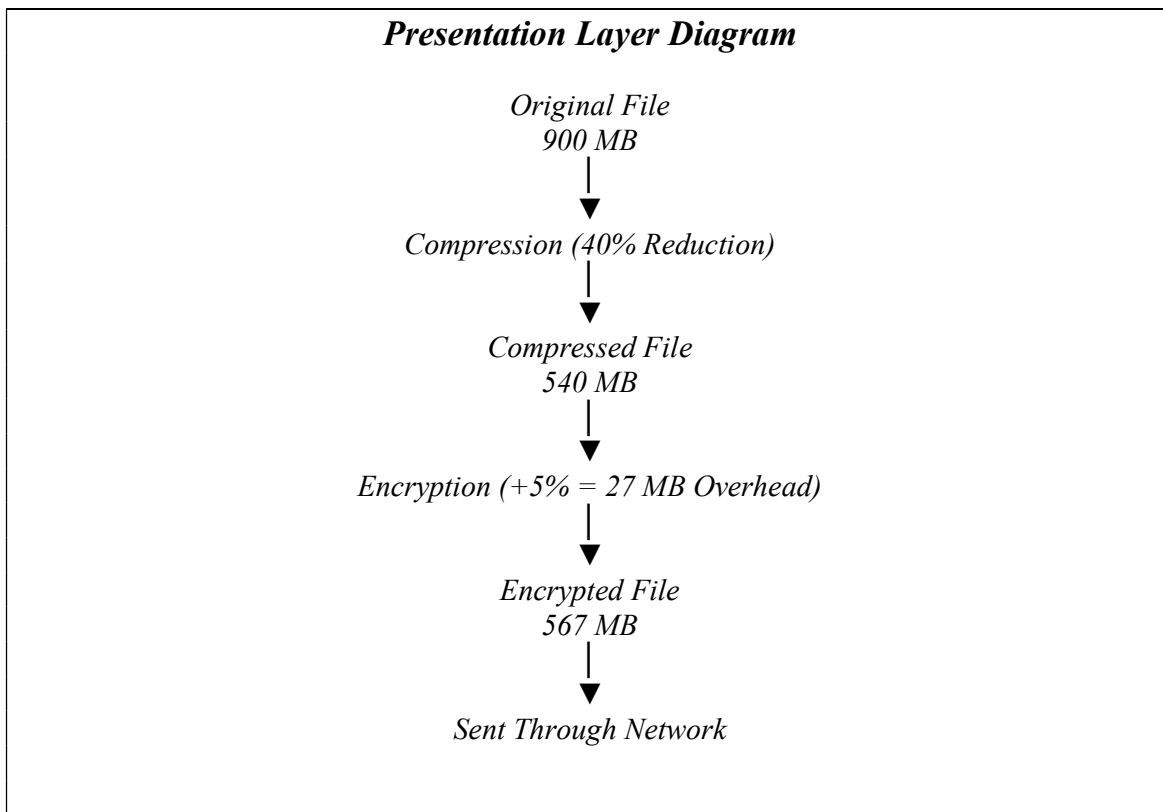
Calculation	Answer
Original File Size	900 MB
Compressed File Size	540 MB
Encrypted File Size	567 MB
Total Percentage Reduction	37%

Presentation Layer Functions

The **Presentation Layer (Layer 6)** is responsible for formatting, compressing, encrypting, and decrypting data before transmission.

Advantages

- *Compresses data to reduce file size.*
- *Encrypts data for secure communication.*
- *Converts data into a standard format.*
- *Improves transmission efficiency.*
- *Protects data from unauthorized access.*



Question:7

An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the **total number of requests generated** and **estimate the total transmission time in seconds**. Discuss how the **Application Layer** provides reliable services to users.

Given Data

- Total Data = **3 GB**
- Throughput = **120 Mbps**
- Request Size = **3 MB**

Step 1: Convert GB to MB

$$1 \text{ GB} = 1000 \text{ MB}$$

$$3 \text{ GB} = 3000 \text{ MB}$$

Answer: 3000 MB

Step 2: Calculate Number of Requests

Formula

$$\begin{aligned} \text{Number of Requests} &= \frac{\text{Total Data}}{\text{Request Size}} \\ &= \frac{3000}{3} = 1000 \end{aligned}$$

Answer: 1000 Requests

Step 3: Calculate Transmission Time

Convert **3 GB** into **Megabits**.

$$3000 \text{ MB} \times 8 = 24000 \text{ Mb}$$

Formula

$$\begin{aligned} \text{Transmission Time} &= \frac{\text{Total Data (Mb)}}{\text{Throughput (Mbps)}} \\ &= \frac{24000}{120} = 200 \text{ seconds} \end{aligned}$$

Answer: 200 Seconds

Final Answers

Calculation	Answer
Total Data	3 GB (3000 MB)
Number of Requests	1000
Transmission Time	200 Seconds

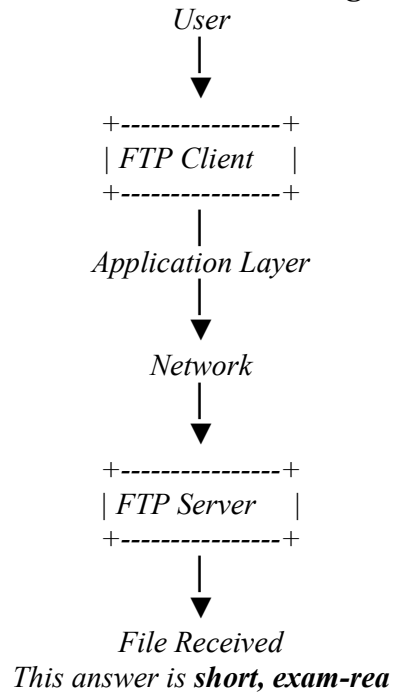
Application Layer Functions

The **Application Layer (Layer 7)** provides network services directly to the user and supports applications such as **FTP, HTTP, SMTP, DNS, and Telnet**.

Advantages

- Provides file transfer services (FTP).
- Allows communication between applications.
- Supports user authentication.
- Ensures reliable file transfer.
- Provides network access to user applications.

FTP Communication Diagram



Question:8

*A packet experiences the following processing delays at different OSI layers: **Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms.** The **Propagation Delay = 18 ms** and **Transmission Delay = 12 ms.** Calculate the **total end-to-end delay** experienced by the packet and explain how each layer contributes to the communication process.*

Given Data

- *Application Layer Delay = 4 ms*
- *Transport Layer Delay = 3 ms*
- *Network Layer Delay = 5 ms*
- *Data Link Layer Delay = 2 ms*
- *Physical Layer Delay = 1 ms*
- *Propagation Delay = 18 ms*
- *Transmission Delay = 12 ms*

Step 1: Calculate Total Processing Delay

Formula

$$\begin{aligned}\text{Processing Delay} &= 4 + 3 + 5 + 2 + 1 \\ &= 15 \text{ ms}\end{aligned}$$

Answer: 15 ms

Step 2: Calculate Total End-to-End Delay

Formula

$$\begin{aligned}\text{Total Delay} &= \text{Processing Delay} + \text{Propagation Delay} + \text{Transmission Delay} \\ &= 15 + 18 + 12 \\ &= 45 \text{ ms}\end{aligned}$$

Answer: 45 ms

Final Answers

Calculation	Answer
<i>Processing Delay</i>	15 ms
<i>Propagation Delay</i>	18 ms
<i>Transmission Delay</i>	12 ms
Total End-to-End Delay	45 ms

Contribution of Each Layer

- **Application Layer:** Creates and processes user data.
- **Transport Layer:** Segments data and provides reliable delivery.
- **Network Layer:** Determines the best route using IP addresses.
- **Data Link Layer:** Frames the data and performs error detection.
- **Physical Layer:** Transmits data as electrical, optical, or radio signals.

End-to-End Delay Diagram

Sender

Application Layer (4 ms)



Transport Layer (3 ms)



Network Layer (5 ms)



Data Link Layer (2 ms)



Physical Layer (1 ms)



Transmission Delay (12 ms)



Propagation Delay (18 ms)



Receiver

Total End-to-End Delay = 45 ms

Question:9

A network transfers **80 MB** of useful data using **frames of 1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the **total number of frames required**, the **total overhead transmitted**, and the **transmission efficiency as a percentage**. Explain how protocol overhead affects network performance.

Given Data

- Useful Data = **80 MB**
- Frame Size = **1600 Bytes**
- Overhead per Frame = **80 Bytes**

Step 1: Convert Useful Data into Bytes

$$\begin{aligned} 80 \text{ MB} &= 80 \times 1,000,000 \\ &= 80,000,000 \text{ Bytes} \end{aligned}$$

Step 2: Calculate Number of Frames

Formula

$$\begin{aligned} \text{Number of Frames} &= \frac{\text{Useful Data}}{\text{Frame Size}} \\ &= \frac{80,000,000}{1600} \\ &= 50,000 \end{aligned}$$

Answer: 50,000 Frames

Step 3: Calculate Total Overhead

Formula

$$\begin{aligned} \text{Total Overhead} &= 50,000 \times 80 \\ &= 4,000,000 \text{ Bytes} \\ &= 4 \text{ MB} \end{aligned}$$

Answer: 4 MB

Step 4: Calculate Transmission Efficiency

Useful data transmitted:

$$80 \text{ MB}$$

Total transmitted data:

$$80 + 4 = 84 \text{ MB}$$

Formula

$$\begin{aligned} \text{Efficiency} &= \frac{80}{84} \times 100 \\ &= 95.24\% \end{aligned}$$

Answer: 95.24%

Final Answers

<i>Calculation</i>	<i>Answer</i>
<i>Number of Frames</i>	50,000
<i>Total Overhead</i>	4 MB
<i>Transmission Efficiency</i>	95.24%

Effect of Protocol Overhead

Protocol overhead is the extra information (headers and trailers) added to each frame for communication.

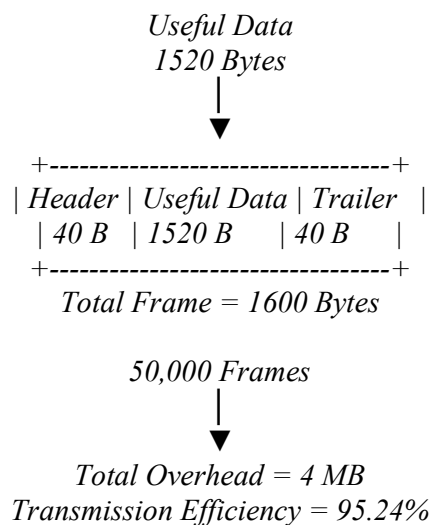
Advantages

- *Identifies sender and receiver.*
- *Detects transmission errors.*
- *Ensures reliable communication.*
- *Supports routing and data delivery.*

Disadvantages

- *Increases total transmitted data.*
- *Consumes additional bandwidth.*
- *Slightly reduces transmission efficiency.*

Protocol Overhead Diagram



Question:10

A hospital transfers a patient's **240 MB** medical record through an **IoT-enabled healthcare network**. The **Presentation Layer** compresses the data by **30%**, the **Transport Layer** divides the compressed data into **1400-byte segments**, and the **Data Link Layer** adds **60 bytes of overhead** to each frame. Calculate the **compressed file size**, the **total number of segments transmitted**, the **total protocol overhead introduced**, and the **final amount of data transmitted across the Physical Layer**. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given Data

- Original File Size = **240 MB**
- Compression = **30%**
- Segment Size = **1400 Bytes**
- Data Link Layer Overhead = **60 Bytes/Frame**

Step 1: Calculate Compressed File Size

Formula

$$\begin{aligned}\text{Compressed Size} &= 240 \times (1 - 0.30) \\ &= 240 \times 0.70 \\ &= 168 \text{ MB}\end{aligned}$$

Answer: 168 MB

Step 2: Convert Compressed File into Bytes

$$168 \times 1,000,000 = 168,000,000 \text{ Bytes}$$

Step 3: Calculate Number of Segments

Formula

$$\begin{aligned}\text{Segments} &= \frac{168,000,000}{1400} \\ &= 120,000\end{aligned}$$

Answer: 120,000 Segments

Step 4: Calculate Total Protocol Overhead

Formula

$$\begin{aligned}\text{Overhead} &= 120,000 \times 60 \\ &= 7,200,000 \text{ Bytes} \\ &= 7.2 \text{ MB}\end{aligned}$$

Answer: 7.2 MB

Step 5: Calculate Final Data Transmitted

Formula

$$\begin{aligned}\text{Final Data} &= 168 + 7.2 \\ &= 175.2 \text{ MB}\end{aligned}$$

Answer: 175.2 MB

Final Answers

<i>Calculation</i>	<i>Answer</i>
<i>Compressed File Size</i>	168 MB
<i>Number of Segments</i>	120,000
<i>Total Protocol Overhead</i>	7.2 MB
<i>Final Data Transmitted</i>	175.2 MB

Working of Multiple OSI Layers

- ***Presentation Layer:*** Compresses and encrypts the medical record for efficient and secure transmission.
- ***Transport Layer:*** Divides the compressed file into smaller segments and ensures reliable delivery.
- ***Data Link Layer:*** Adds headers and trailers for framing and error detection.
- ***Network Layer:*** Routes the data through the network using IP addresses.
- ***Physical Layer:*** Transmits the data as electrical, optical, or wireless signals.

*These layers work together to ensure that the patient's medical record is transmitted **securely, accurately, and efficiently.***

